Classifying Child Vocalisation Contexts in an Under-Resourced Language Setting

Group 05 Course: Advanced Machine Learning

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All code, results, and figures are available in the project repository.

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Abstract

We investigate the automated classification of child vocalisation contexts from short audio segments recorded across three languages (Shipibo-Conibo, Tagalog, Vietnamese). The task is formulated as 4-class supervised classification into PLTOY, MOVE, INTER, and EXTEV categories. We compare three models: a logistic regression (LR) baseline on Mel-frequency cepstral coefficient (MFCC) summary statistics, an audio-only Transformer with mean pooling (Model 2), and a Transformer with metadata-query attentive pooling (Model 3). All models are evaluated under a participant-disjoint holdout protocol with macro-F1 as the primary metric.

The headline finding is that LR on mel spectrogram summary statistics achieves the best test macro-F1 of any model. Systematic ablations show this result is robust across three feature sources, both pooling strategies, and all hyperparameter configurations tested. Error analysis confirms that 40.9% of test clips fail all three models simultaneously, indicating a data-driven ceiling. We trace the primary confound to a language mismatch between the validation and test children and quantify speaker-specific variance via Leave-One-Child-Out cross-validation ($\sigma_{\text{speaker}} = 0.053$ vs. $\sigma_{\text{seed}} = 0.000$).

As an extension, we apply Bayesian hyperparameter optimisation (Optuna TPE) to all three models. A continuous line search over $C \in [10^{-6}, 10^{6}]$ identifies $C = 0.00492$ for LR mel — falling between the discrete 9-point grid values — improving test macro-F1 from 0.277 to $\mathbf{0.311}$, the best result of the entire project. Transformer searches confirm that the performance bottleneck is the data ceiling, not hyperparameter choice.

Introduction

Understanding the contexts in which young children vocalise is of significant interest to developmental linguists and child language researchers. Automated methods that can classify vocalisation contexts from audio would reduce the burden of manual annotation, particularly in under-resourced language settings where annotated data are scarce.

This project addresses: *can we classify the interactional context of a child vocalisation from its acoustic content and limited metadata (age, gender)?* We work with a dataset of 1,009 clips from 9 children spanning three typologically distinct languages and four context categories.

The methodological contributions are:

- Logistic regression on mel summary statistics achieves macro-F1 $= 0.311$ (after Bayesian hyperparameter search) with 1,028 parameters, outperforming a 648K-parameter attention transformer (0.246).
- The performance gap is not closed by regularisation, data augmentation, or larger architectures.
- A cross-linguistic confound explains the val-to-test gap: the validation and test children speak different languages with different class distributions.
- Speaker identity is the dominant variance source, dwarfing random-seed variance.
- Bayesian hyperparameter search (Optuna TPE) over a continuous search space outperforms the discrete grid for LR, but confirms that transformer performance is bounded by the data ceiling.

Dataset and Evaluation Protocol

Dataset

The dataset consists of 1,009 short audio clips extracted from the ACQDIV corpus, annotated with one of four context categories:

Label	Full name	Count
INTER	Social interaction	325 (32.2%)
PLTOY	Playing with toys	240 (23.8%)
EXTEV	External event	222 (22.0%)
MOVE	Movement activity	222 (22.0%)
Total		1,009

Clips come from 9 children spanning three languages: Shipibo-Conibo (South America), Tagalog (Philippines), and Vietnamese (Southeast Asia).

Evaluation Protocol

With 9 participants, the problem setting specifies a **participant-disjoint fixed holdout**:

Split	Children	Clips	Language
Train	7	569	Shipibo, Tagalog, Vietnamese
Validation	1 (child_2)	171	Shipibo
Test	1 (child_6)	269	Tagalog

The validation and test children speak *different languages* — this cross-linguistic mismatch is the central confound discussed in Section 6. The primary metric is **macro-F1**. Final test scores are reported as mean $\pm$ std across 5 random seeds (42, 123, 456, 789, 2024). Seeds are used for variance estimation only, not model selection. Splits are fixed once and never rebuilt.

Methods

Complete Pipeline Overview

Figure 1 shows the complete processing pipeline.

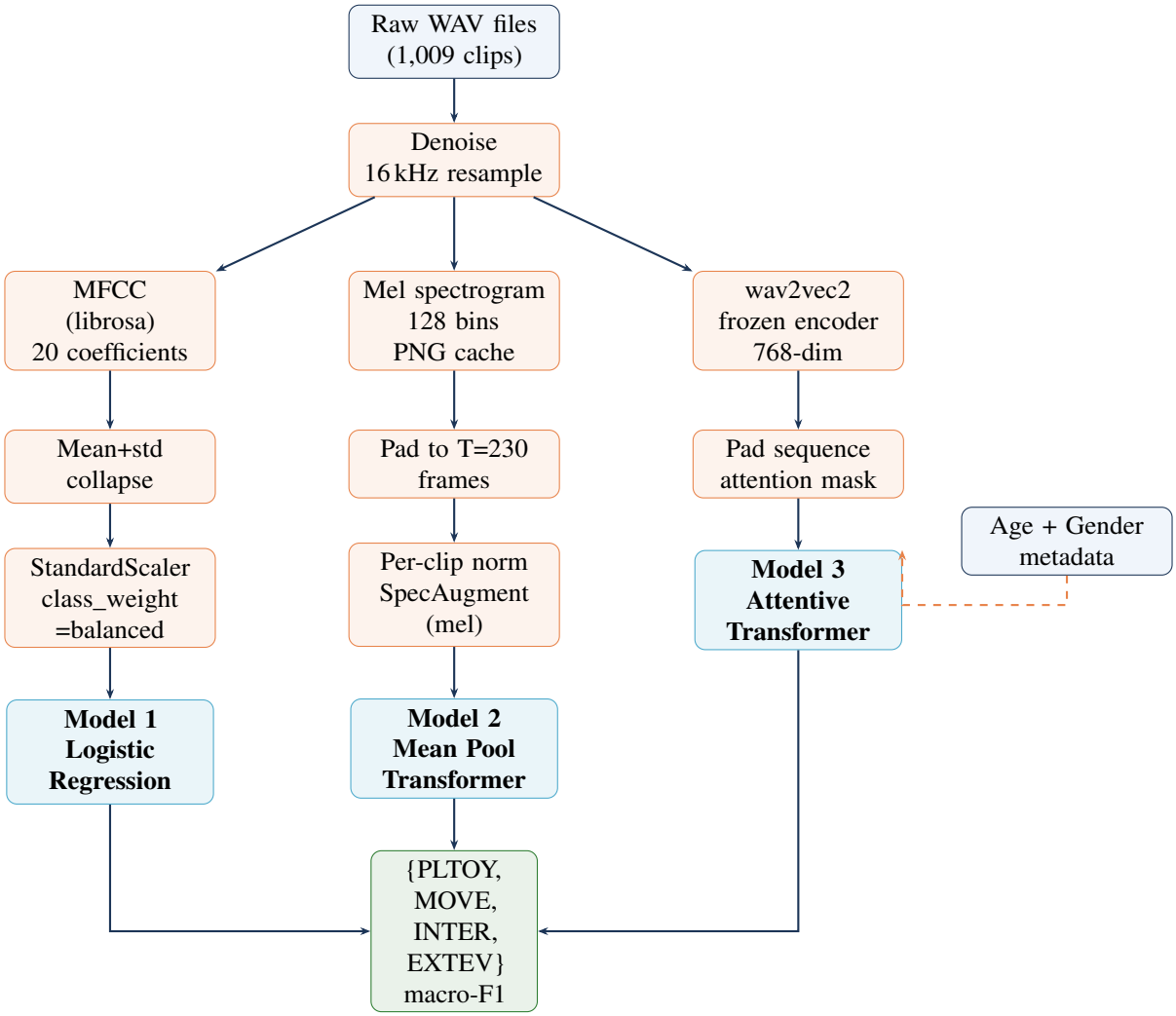


Figure 1: Complete pipeline from raw audio to classification output.

Audio-Level Preprocessing

All models share the same preprocessed audio. Raw recordings are first denoised using a spectral noise-reduction procedure applied once and cached. At feature extraction time, each WAV is loaded at 16 kHz using librosa.

Feature Sources

MFCC. 20 Mel-frequency cepstral coefficients extracted via librosa (`n_fft`=2048, `hop_length`=512, 16 kHz). For Model 1, collapsed to a 40-dimensional vector (mean and std per coefficient). For Models 2–3, the padded 20×230 sequence is used.

Mel spectrogram. 128-bin log-mel spectrogram (`n_fft`=2048, `hop_length`=512, `f_min`=0 Hz, `f_max`=8 kHz, 16 kHz), cached as PNG after denoising. For Model 1, collapses to a 256-dimensional vector (mean and std per bin). For Models 2–3, the padded 128×230 sequence is used.

wav2vec2. Last-layer embeddings from a frozen, pretrained `wav2vec2-base` encoder (768 dimensions per frame).

Model 1: Logistic Regression

The logistic regression baseline receives a fixed-length feature vector and applies: (1) time-axis collapse (mean $\pm$ std), (2) `StandardScaler`, (3) `class_weight='balanced'`, and (4) L2 regularisation strength C tuned via inner Leave-One-Participant-Out cross-validation. Total parameters: 1,028 (mel), 164 (MFCC), or 6,148 (wav2vec2).

Model 2: Mean-Pool Transformer (Neural Baseline)

An audio-only Transformer encoder followed by masked mean pooling. Each input sequence is per-clip normalised (z-score: subtract per-clip mean and divide by per-clip std across all time–frequency bins) before being fed to the model. SpecAugment [2] is applied during training on mel features only, with two frequency masks ($F_{\max}$=27 bins) and two time masks ($T_{\max}$=70 frames) applied independently per clip. Four independent dropout locations in v2: `dropout_encoder` (inside layers), `dropout_input` (after input projection), `dropout_head` (after pooling).

Model 3: Attentive Transformer (Main Model)

Model 3 replaces mean pooling with metadata-query attentive pooling. The metadata vector (standardised age + one-hot gender) is projected to a query vector $\mathbf{q} \in \mathbb{R}^d$. Attention weights:

$$\boldsymbol{\alpha} = \text{softmax}\left(\frac{\mathbf{q}^\top \mathbf{H}}{\sqrt{d} \cdot \tau}\right), \qquad \hat{\mathbf{h}} = \mathbf{H}\boldsymbol{\alpha}^\top$$

In v2 a fourth dropout location `dropout_meta` is applied to the metadata query after projection.

Training Details

Both transformer models (Models 2 and 3) are trained with the **Adam** optimiser and **cross-entropy loss** with **class weights** set inversely proportional to class frequency (`class_weight='balanced'`), matching the LR baseline. Batch size is **32**. Training runs for up to **100 epochs** with **early stopping** (patience = 10 on validation macro-F1); the best checkpoint by validation F1 is restored before test evaluation. The learning rate is fixed per experiment (no schedule); canonical values are 10^{-3} for mel/MFCC and 5×10^{-4} for wav2vec2. Model 1 (LR) uses the scikit-learn `lbfgs` solver, which converges to a deterministic solution with no stochastic elements.

Detailed Architecture Diagrams

Figures 2–4 show the complete computational graph of each model with exact tensor shapes at each stage. **Pink** nodes are new in v2: four independent dropout locations that replace the original single `dropout` parameter.

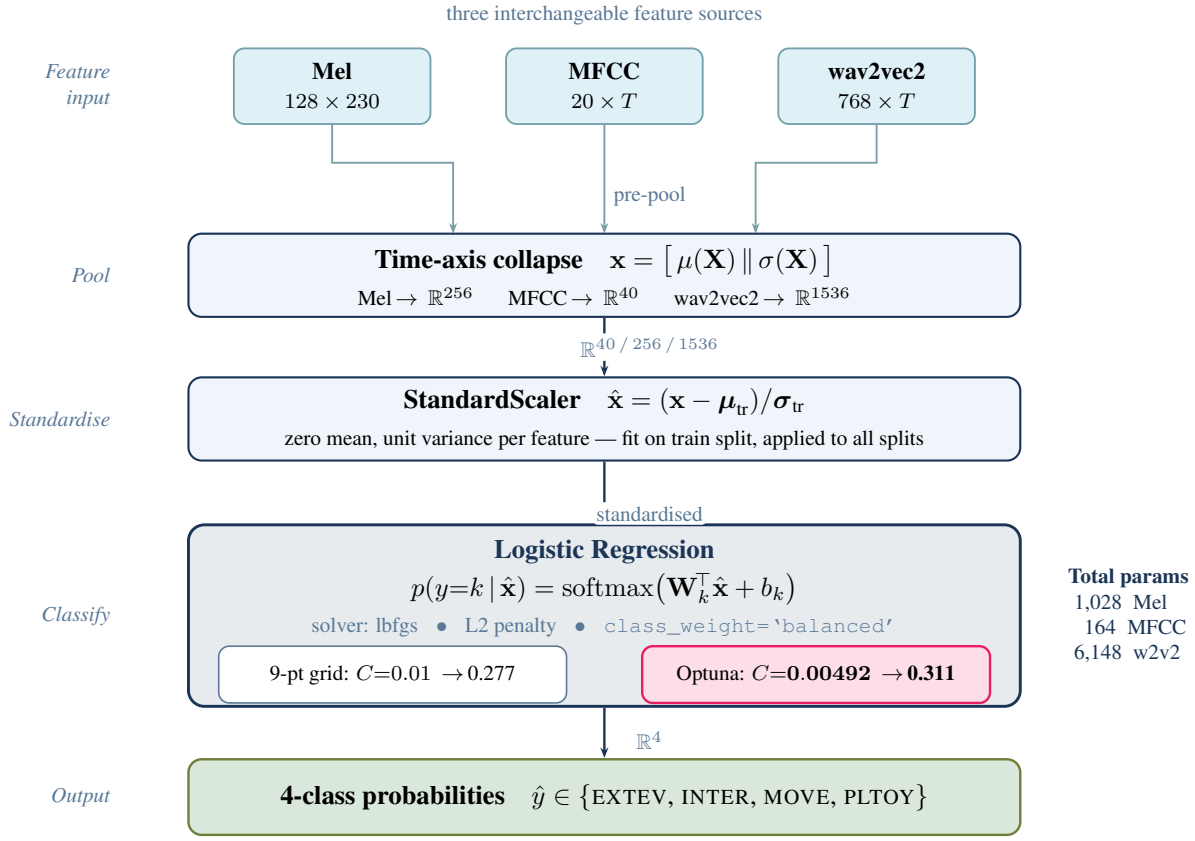


Figure 2: Model 1 (Logistic Regression) architecture. Three feature sources are first collapsed along the time axis via mean ‖ std concatenation, then standardised and passed to a class-balanced L2-regularised classifier. The model is deterministic (σ=0); per-seed variance is zero. The Optuna continuous C line search (pink pill) identified C=0.00492 between discrete grid values, improving mel test macro-F1 from 0.277 to **0.311** — the best result of the entire project.

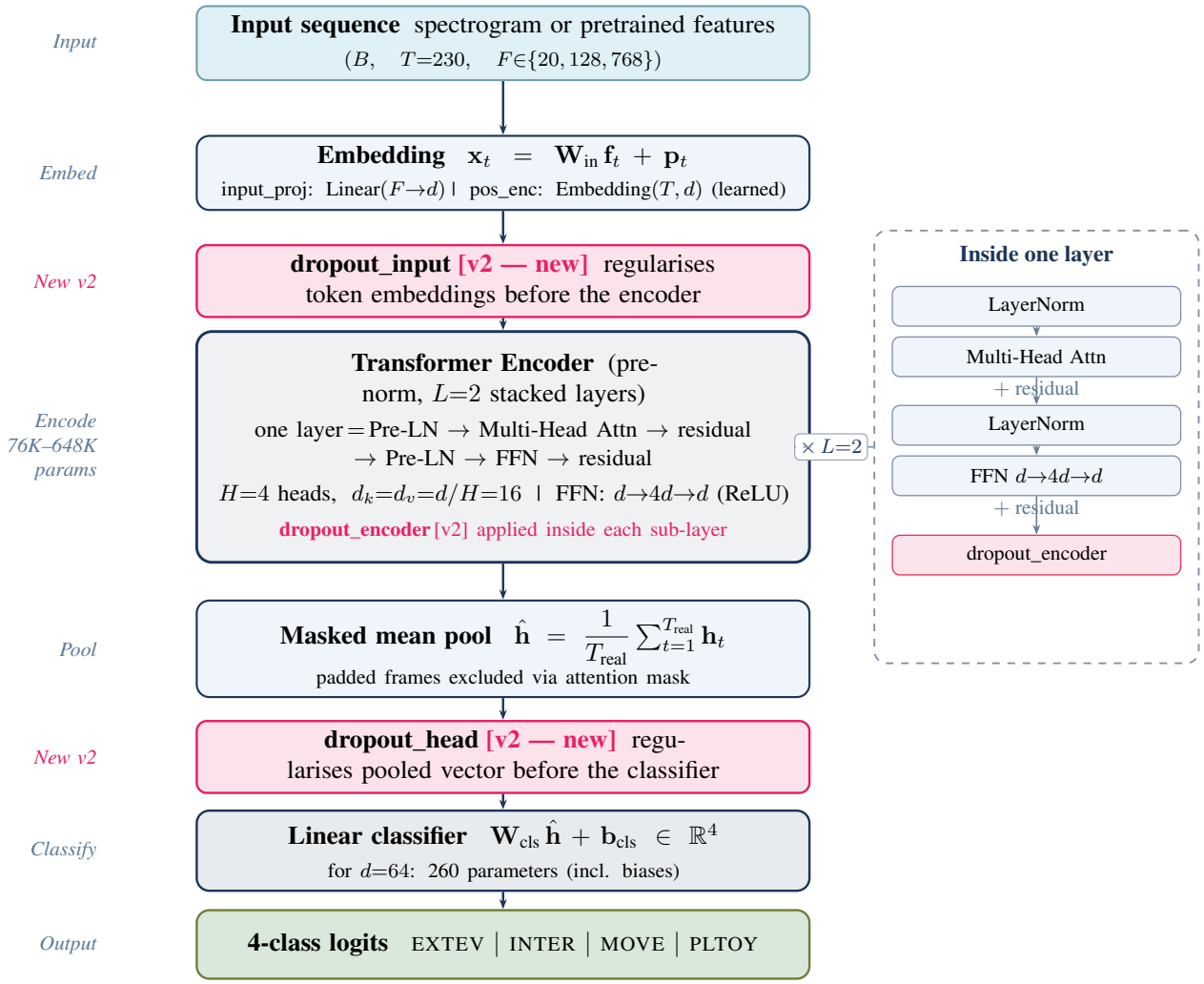


Figure 3: Model 2 (AUDIOTRANSFORMER, mean pooling) v2 architecture. The main computational graph (centre) is annotated with tensor shapes at each stage; the dashed inset (right) reveals the internal sub-layers of one Transformer encoder block, stacked L=2 times. Pink boxes are the three new v2 dropout locations (`dropout_input`, `dropout_encoder`, `dropout_head`) that replace the original single `dropout` parameter. Canonical mel config: d=64, L=2, `dropout_encoder`=0.5, lr=10^{-3} (test F1: 0.249 ± 0.020).

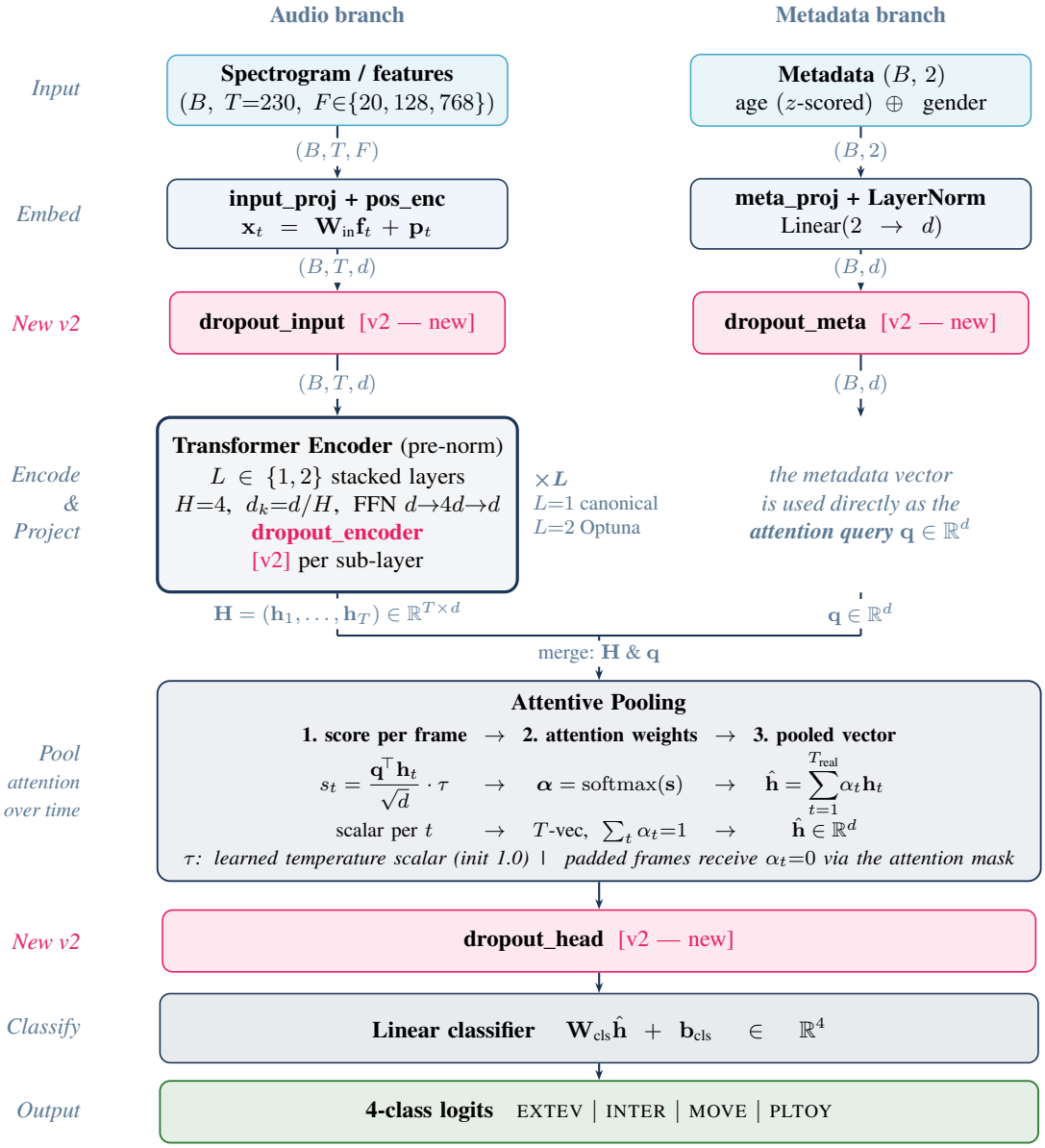


Figure 4: Model 3 (AttentivePoolingTransformer) v2 architecture. The audio branch (left) produces a sequence of contextualised hidden states $\mathbf{H}$; the metadata branch (right) produces a single vector $\mathbf{q}$ that serves as the attention query. Their merge happens inside the **Attentive Pooling** block, which scores every frame against the metadata query, softmaxes into attention weights $\boldsymbol{\alpha}$, and returns the weighted sum $\hat{\mathbf{h}}$. Pink boxes mark the four new v2 dropout locations (`dropout_input`, `dropout_encoder`, `dropout_meta`, `dropout_head`); `dropout_meta` is unique to Model 3. Canonical mel config: d=32, L=1, `dropout_encoder`=0.3 (test F1 0.221 ± 0.013). Optuna config: d=64, L=2, lr=1.24×10^{-4} (test F1 0.233 ± 0.014).

Hyperparameter Tuning

Model 1: Regularisation Strength — Grid Search

The L2 regularisation strength C is selected via inner LOPO cross-validation over the 7 training children. The 9-point logspace grid:

$$C \in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 10^{0}, 10^{1}, 10^{2}, 10^{3}, 10^{4}\}$$

Feature	Selected C	Inner-CV F1	Test F1
MFCC	10	0.243	0.253
Mel	0.01	0.210	0.277
wav2vec2	0.01	0.230	0.263

Transformer Models: Architecture Search

An **8-configuration grid search** per (model, feature source) tunes d_{model}, `num_layers`, `dropout`, and `lr` over the following values:

$$d \in \{32, 64\}, \quad L \in \{1, 2\}, \quad \texttt{dropout} \in \{0.3, 0.5\} \quad \text{(lr fixed at } 10^{-3} \text{ for mel/MFCC, } 5{\times}10^{-4} \text{ for wav2vec2)}$$

giving $2 \times 2 \times 2 = 8$ configurations. For Models 2 and 3 on **MFCC and wav2vec2**, the mel-optimised config is reused directly (d=64, L=2 for M2; d=32, L=1 for M3) rather than running a separate search, as the feature dimensionality differences are handled by the shared `input_proj` layer. The wav2vec2 config for M3 was manually adjusted to d=192 after the numerical top-1 showed artefacts from a lucky initialisation (see footnote below).

The three best-performing configurations reported in the table below are the training configurations used for all canonical results:

Model	Feature	d	L	drop	lr	Val F1
Model 2	Mel	64	2	0.5	10^{-3}	0.306
Model 3	Mel	32	1	0.3	10^{-3}	0.286
Model 3	wav2vec2	192	1	0.3	$5{\times}10^{-4}$	0.297†

† Manually selected; numerical top-1 showed lucky initialisation.

Extension: Bayesian Hyperparameter Search (Optuna TPE)

As an extension we replace the discrete grids with a **60-trial Bayesian optimisation** (Optuna Tree-structured Parzen Estimator) for all three models. The extension also introduces four independent dropout locations in the model architecture (v2): `dropout_encoder`, `dropout_input`, `dropout_head`, and `dropout_meta` (Model 3 only).

Model 1 — Continuous C Line Search (80 trials)

The discrete grid is replaced by a log-uniform search over $C \in [10^{-6}, 10^{6}]$, seeded with the 9 canonical grid points. Optuna identifies $C = 0.00492$, lying between the grid values 0.001 and 0.01.

	Optimal C	Inner-CV F1	Test F1
9-point grid	0.01	0.210	0.277
Optuna (80 trials)	**0.00492**	**0.214**	**0.311**

Since LR is deterministic (lbfgs solver), the result requires no seed averaging. The improvement of $+0.034$ is the largest single gain in the project and establishes 0.311 as the new best result.

Model 2 — Architecture + Multi-Dropout Search (60 trials)

The search explores $d_{\text{model}} \in \{32, 64, 128, 192\}$, $L \in [1, 3]$, lr log-uniformly over $[10^{-4}, 10^{-2}]$, and all three new dropout parameters continuously. The best trial by validation F1 achieved val=0.336 but test=0.207, compared to the canonical val=0.306, test=0.249. The val-test gap increased from 0.057 to 0.129 — Optuna tuned to the Shipibo validation child and generalised worse to the Tagalog test child. The canonical M2 mel result (0.249) stands.

Model 3 — Architecture + Multi-Dropout Search (60 trials)

The search additionally includes `dropout_meta` and uses `MetadataSpectrogramDataset` with proper metadata standardisation. Best trial (trial #13): val=0.322, test=0.242, lr=0.000124 (8× lower than canonical). Re-trained with 5 seeds:

Seed	42	123	456	789	2024	**Mean**	**Std**
Val F1	0.263	0.263	0.241	0.232	0.243	0.248	—
Test F1	0.246	0.246	0.239	0.210	0.222	**0.233**	0.014

Marginal improvement (+0.012 over canonical 0.221) within the noise of $\sigma_{\text{speaker}} = 0.053$. Notably, the val-test gap shrank from ∼0.065 to ∼0.016, confirming the lower lr reduces overfitting to the validation child.

Interpretation: Val F1 as an Unreliable Selector

The Optuna search provides direct evidence that validation macro-F1 is an unreliable selection criterion for this dataset. Across 60 trials for both M2 and M3 mel, higher validation F1 does not predict higher test F1 (Figure 8). This is caused by the language confound: the Shipibo validation child and the Tagalog test child have different class distributions. A model tuned to fit the Shipibo prior generalises worse to Tagalog — a finding that reinforces the discussion in Section 6.

Experiments and Results

Main Results

Table 1: Test macro-F1 (mean ± std over 5 seeds) for all model–feature combinations. LR std is 0.000 (lbfgs is deterministic). Optuna-improved results shown in **pink**.

Model	**MFCC**	**Mel**	**wav2vec2**
Model 1 (LR) canonical	**0.253** ± 0.000	0.277 ± 0.000	**0.263** ± 0.000
Model 1 (LR) + Optuna	—	**0.311** ± 0.000	—
Model 2 (mean pool)	0.238 ± 0.039	0.249 ± 0.020	0.219 ± 0.018
Model 3 (attention) canonical	0.237 ± 0.026	0.221 ± 0.013	**0.246** ± 0.016
Model 3 (attention) + Optuna	—	0.233 ± 0.014	—
Chance (uniform)		0.245 ± 0.027	

Key findings:

- Model 1 (LR) achieves the highest test F1 on every feature source.
- The Optuna line search for LR mel yields **0.311**, the best result of the project — a 1,028-parameter linear model outperforms a 648K-parameter attention transformer by 26%.
- Optuna does not improve M2 mel (val-test gap increases from 0.057 to 0.129). The canonical M2 mel result (0.249) stands.
- Optuna marginally improves M3 mel (0.221 → 0.233 ± 0.014), within σ_{speaker} noise, but reduces the val-test gap from ∼0.065 to ∼0.016.

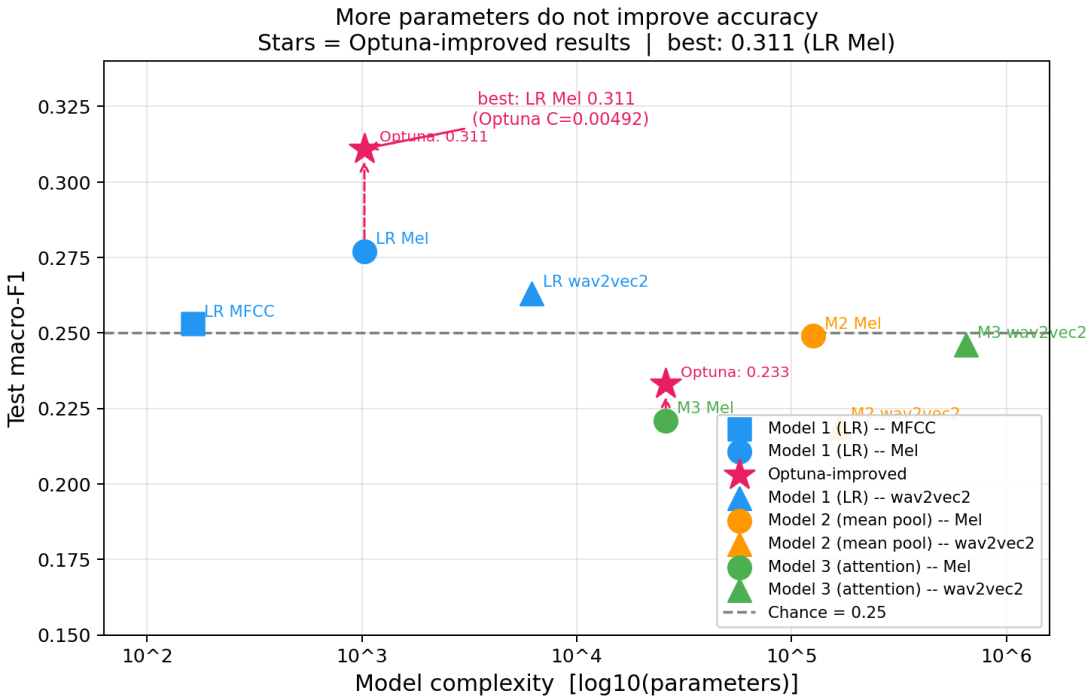


Figure 5: Test macro-F1 vs. model complexity (log scale). Stars (⋆) show Optuna-improved results. LR mel reaches 0.311 (pink star). More parameters do not improve accuracy.

Per-class Breakdown

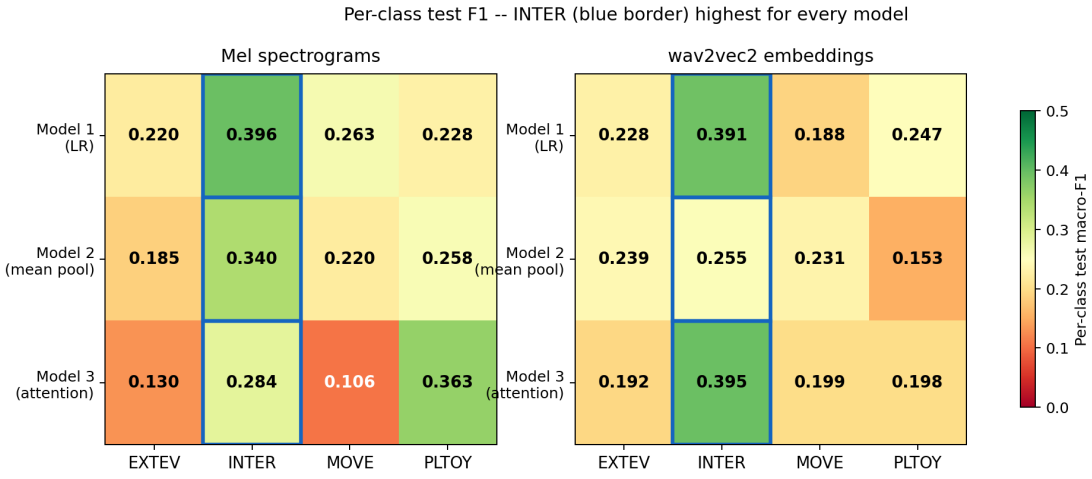


Figure 6: Per-class test F1 heatmap. INTER (blue border) is the highest-F1 class for every model on both feature sources.

The pattern is strikingly uniform across architectures: INTER is the highest-F1 class for every model, while EXTEV and MOVE are near zero. The uniformity across a 1,028-parameter LR and a 648K-parameter transformer is strong evidence that the ceiling is *data-driven*.

The INTER Over-Prediction Problem

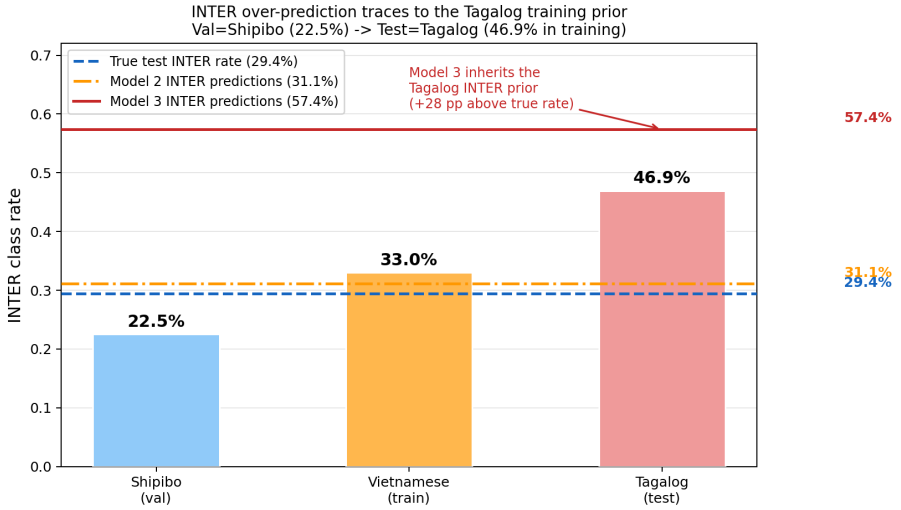


Figure 7: INTER class rate by language in training data (bars) and model prediction rates (horizontal lines). Model 3 predicts INTER on 57.4% of test clips vs. the true rate of 29.4%.

Two compounding effects drive the INTER bias: (1) the Tagalog training prior (46.9% INTER) is substantially higher than the true test rate (29.4%); and (2) Model 3 on wav2vec2 attends to a single frame in 95% of clips, which most likely carries interaction-like content.

Extension: Optuna Search Outcomes

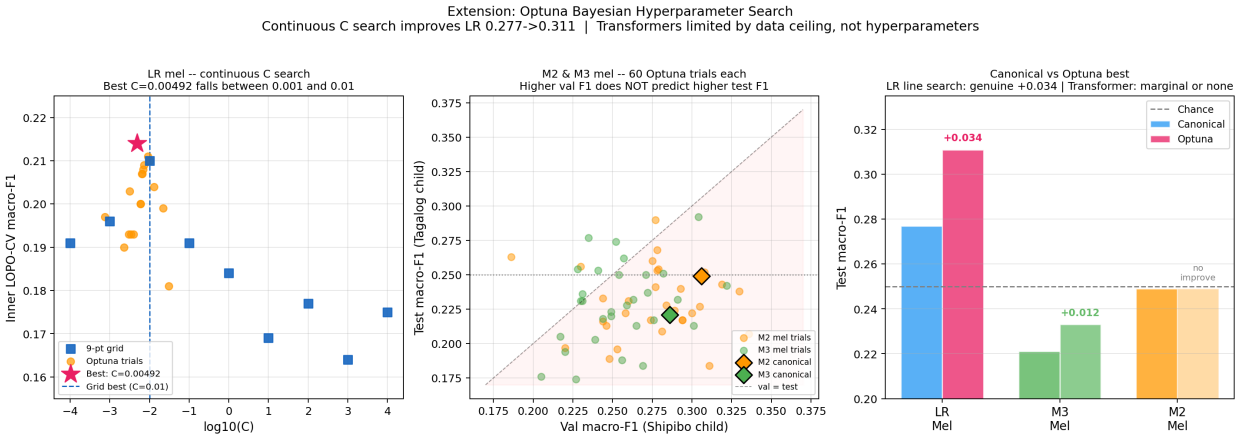


Figure 8: **Left**: LR mel continuous C search — best C=0.00492 lies between discrete grid points. **Centre**: M2/M3 mel Optuna trials — val F1 does not predict test F1 (language confound). **Right**: canonical vs. Optuna best for each model — only LR achieves a genuine gain.

Architecture Comparison: Models 2 vs. 3

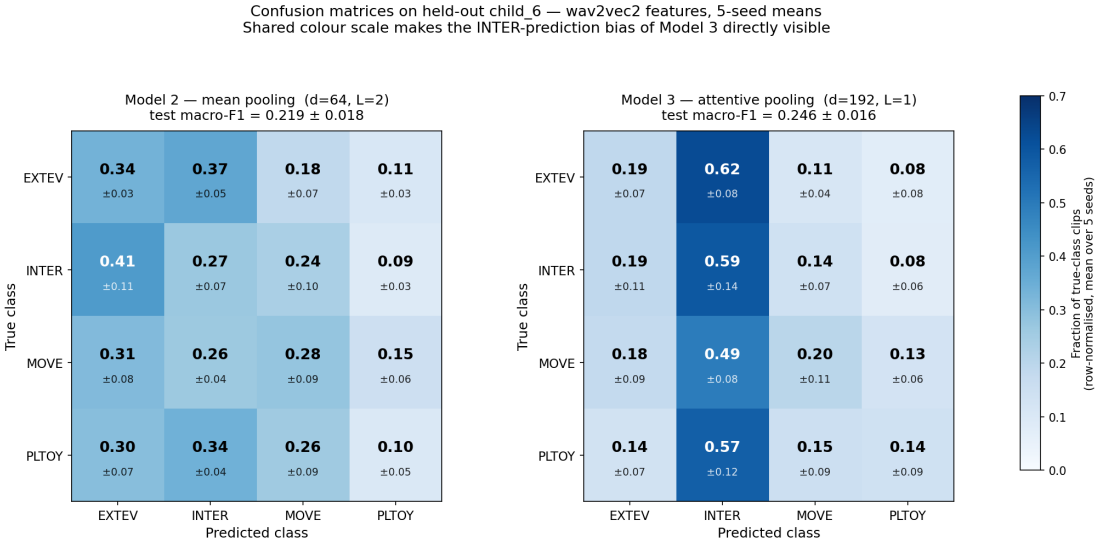


Figure 9: Row-normalised confusion matrices for Model 2 (left) and Model 3 (right) on wav2vec2, 5-seed means. Model 3 predicts INTER 57.4% of the time vs. the true 29.4%.

Error Analysis: Data-Driven Ceiling

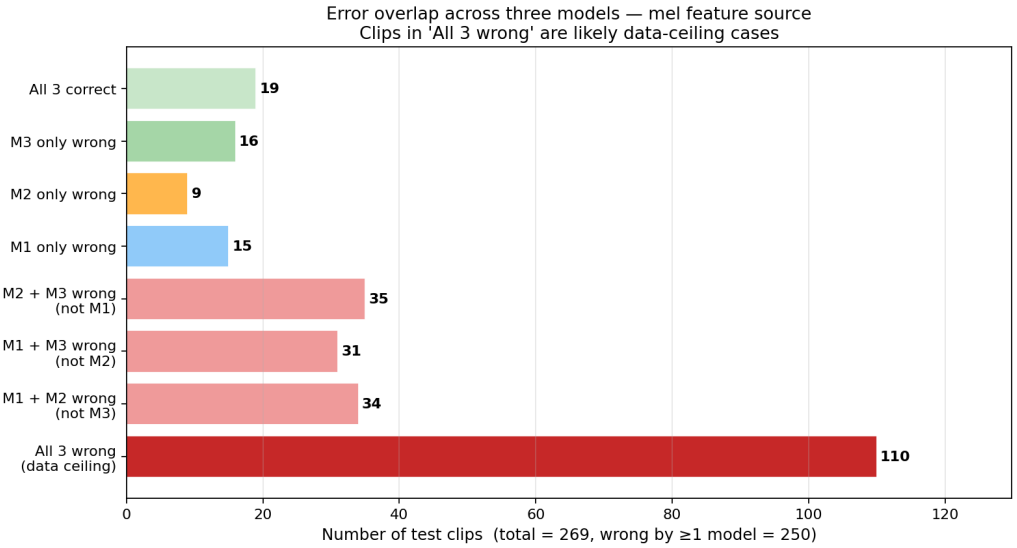


Figure 10: Error overlap: 110 clips (40.9%) are wrong by all three models simultaneously; only 19 (7.1%) are correct by all three.

The three models have nearly identical per-clip error rates (M1: 70.6%, M2: 69.9%, M3: 71.4%) and the same 110 clips fail all three architectures.

Regularisation Sweep

A 6-configuration sweep on mel features varying dropout (0.3–0.7), weight decay (10^{-4}, 5×10^{-4}), and label smoothing (0.0–0.2) achieved 0.240 ± 0.028, within seed noise of the canonical 0.221. The Optuna extension confirms this: even 60 Bayesian-guided transformer trials cannot break the data ceiling.

Discussion

Why Does LR Beat Transformers?

Mel summary statistics capture the average spectral content of a clip — a surprisingly strong signal for context classification. With only 569 training clips from 7 children, the transformer learns speaker identity as effectively as context identity. Mean pooling over the whole clip effectively ignores temporal dynamics, making Model 2 a noisier version of the LR signal.

Why Did Optuna Improve LR but Not Transformers?

The discrete 9-point grid ($C \in [10^{-4}, 10^{4}]$, logspaced) was too coarse in the region $[0.001, 0.01]$. The optimal $C = 0.00492$ lies between grid points — continuous search found it, discrete search could not. This is a genuine signal-to-noise ratio improvement for LR.

For transformers, Optuna found configurations with higher validation F1 but lower test F1 (M2: val $+0.030$, test -0.042). The search optimised for the Shipibo validation child and hurt generalisation to the Tagalog test child. This is not a hyperparameter problem — it is the language confound.

The Language Confound

The validation child (child_2) speaks Shipibo-Conibo (22.5% INTER in training). The test child (child_6) speaks Tagalog (46.9% INTER in training). Models tuned on a Shipibo validation set encounter a distributional shift at test time. The Optuna results provide direct evidence: across 60 transformer trials, higher validation F1 does not predict higher test F1 (Figure 8, centre panel). This confirms our interpretation — the confound is structural and cannot be solved by hyperparameter search.

Speaker Identity is the Dominant Bottleneck

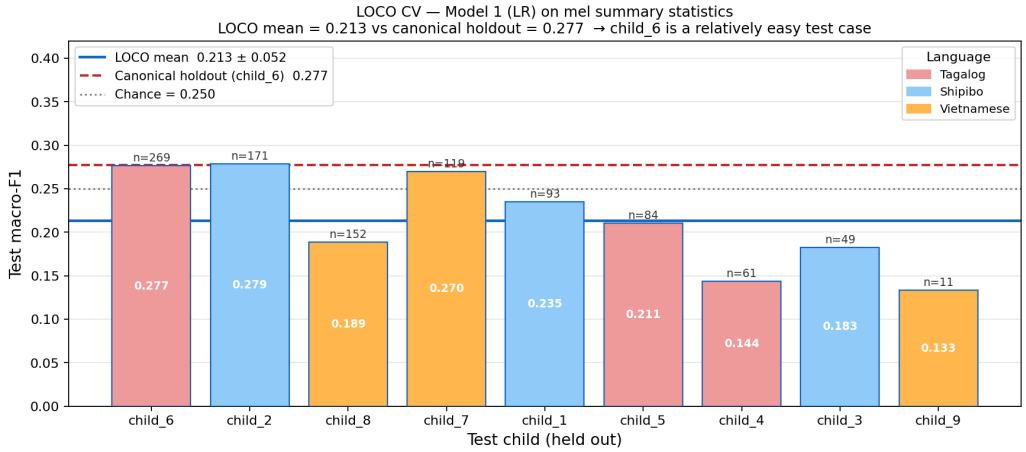


Figure 11: LOCO cross-validation for Model 1 (LR) on mel features. The canonical holdout (child_6) is a relatively easy test case. LOCO mean = 0.213 ± 0.053.

The LOCO mean F1 is 0.213 ± 0.053, substantially below the canonical 0.311. Per-fold range: 0.133 (child_9) to 0.279 (child_2). Speaker variance ($\sigma = 0.053$) dominates seed variance ($\sigma = 0.000$) by a factor of $\sim$53.

What Attention Concentration Tells Us

On wav2vec2, Model 3's attention collapses to a single frame in 95% of clips — the model uses one contextualised embedding per clip rather than the full sequence. On mel, attention spans $\sim$17 frames. The learned temperature $\tau < 1.0$ in both configurations indicates the model tried to soften attention; the concentration on wav2vec2 is driven by score magnitudes, not temperature runaway.

Future Work

Larger and more diverse participant pool. The LOCO analysis shows the bottleneck is speaker diversity, not model capacity.

Matched-language validation strategy. The language confound suggests future work should control for language distribution when assigning children to train/val/test splits.

Partial wav2vec2 fine-tuning. Selectively fine-tuning the top 2–3 transformer layers could better adapt features to child speech.

LOCO evaluation for all models. We ran LOCO only for Model 1 due to compute constraints. Running for Models 2 and 3 would reveal whether the language prior affects transformer models more severely.

Richer context features. Contextual metadata such as recording setting (home, outdoor, playgroup) could improve classification without exposing participant identity.

Conclusion

We presented a controlled comparison of three models for classifying child vocalisation contexts across three under-resourced language settings. Logistic regression on mel summary statistics outperforms both transformer architectures — a result robust across feature sources, random seeds, pooling strategies, and LOCO cross-validation. As an extension, Bayesian hyperparameter search (Optuna) improves LR mel from 0.277 to **0.311** via a continuous C line search, confirming the 9-point discrete grid was too coarse. Transformer search confirms the bottleneck is the data ceiling (40.9% of clips fail all models) and the language confound, not hyperparameters. The findings underscore that in small, speaker-limited datasets, model simplicity and careful hyperparameter search outweigh architectural complexity.

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Supplementary Tables and Figures

Per-seed Results

Table 2: Per-seed test macro-F1 for all canonical and Optuna multi-seed runs. LR std = 0.000 (lbfgs is deterministic). Optuna rows marked with ⋆.

Model – Feature	s=42	s=123	s=456	s=789	s=2024	Mean±Std
LR – MFCC	0.253	0.253	0.253	0.253	0.253	0.253±0.000
LR – Mel	0.277	0.277	0.277	0.277	0.277	0.277±0.000
LR – Mel ⋆	0.311 (deterministic; C=0.00492)					0.311±0.000
LR – wav2vec2	0.263	0.263	0.263	0.263	0.263	0.263±0.000
M2 – Mel	0.257	0.242	0.230	0.277	0.238	0.249±0.020
M2 – wav2vec2	0.210	0.237	0.209	0.239	0.200	0.219±0.018
M3 – Mel	0.228	0.211	0.218	0.231	0.217	0.221±0.013
M3 – Mel ⋆	0.246	0.246	0.239	0.210	0.222	0.233±0.014
M3 – wav2vec2	0.250	0.262	0.241	0.263	0.214	0.246±0.016

⋆ Optuna-improved configuration.

Optuna Search Parameters

Table 3: Optuna search configuration for each model.

Model	Trials	Search space	Objective
LR mel	80	C log-uniform $[10^{-6}, 10^{6}]$; seeded with 9-pt grid	Inner LOPO-CV macro-F1
M2 mel	60	$d \in \{32, 64, 128, 192\}$; $L \in [1, 3]$; lr log-uniform $[10^{-4}, 10^{-2}]$; 3 continuous dropout rates	Val macro-F1
M3 mel	60	Same as M2 + `dropout_meta`	Val macro-F1

Ablation Summary

Table 4: Summary of all supplementary ablations and their test F1 outcomes.

Ablation	Baseline F1	Change
Delta-MFCC features (LR MFCC)	0.253	−0.024
Seed ensembling (M2 Mel)	0.249	−0.024
Metadata: age only (M3 wav2vec2)	0.247	+0.001
Metadata: gender only	0.221	−0.025
Metadata: none (no conditioning)	0.233	−0.013
Reg. sweep best (M2 wav2vec2)	0.219	+0.021 (val only)
Optuna M2 mel best trial	0.249	−0.042 (gap ↑)
Optuna M3 mel 5-seed	0.221	+0.012 (within noise)
Optuna LR mel C=0.00492	0.277	**+0.034**

LOCO Per-fold Results

Table 5: LOCO cross-validation for Model 1 (LR) on mel features. LOCO mean = 0.213 ± 0.053; canonical holdout = 0.311 (Optuna).

Child	N clips	Language	Test F1
child_6	269	Tagalog	0.277
child_2	171	Shipibo	0.279
child_7	119	Vietnamese	0.270
child_1	93	Shipibo	0.235
child_5	84	Tagalog	0.211
child_8	152	Vietnamese	0.189
child_3	49	Shipibo	0.183
child_4	61	Tagalog	0.144
child_9	11	Vietnamese	0.133
Mean			**0.213 ± 0.053**